

PANDAS & MATPLOTLIB

Pandas

- Pandas is a Python library used for working with data sets.
- It has supports for analyzing, cleaning, exploring, and manipulating data.
- The name "Pandas" has a reference to both "Panel Data", and "Python Data Analysis".
- Pandas allows us to analyze big data and make conclusions based on statistical theories.
- Pandas can clean messy data sets, and make them readable and relevant.

Using Pandas

```
C:\Users\Your Name>pip install pandas
```

```
# check pandas version
import pandas
print(pandas.__version__)
```

```
# alternative of above code
import pandas as pd
print(pd.__version__)
```

Pandas DataFrames

- Data sets in Pandas are usually multi-dimensional tables, called **DataFrames**.
- Series is like a column, a DataFrame is the whole table.
- A **Pandas DataFrame** is a 2-dimensional data structure, like a 2-dimensional array, or a table with rows and columns.

DataFrames

```
import pandas as pd
```

```
data = {  
    "calories": [420, 380, 390],  
    "duration": [50, 40, 45]  
}
```

```
# create a DataFrame object from dictionary
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

- Dictionary Keys will be column title and Dictionary Values will be inserted as the column values.
- Rows of the DataFrame can be accessed using index starting from 0.

Output:

	calories	duration
0	420	50
1	380	40
2	390	45

DataFrames

```
import pandas as pd

data = {
    "calories": [420, 380, 390],
    "duration": [50, 40, 45]
}

#load data into a DataFrame object:
df = pd.DataFrame(data)

print(df)
#use a row index:
print(df.loc[0])
#use a list of indexes:
print(df.loc[[0, 2]])
```

Output:

	calories	duration
0	420	50
1	380	40
2	390	45

calories	420
duration	50

Name: 0, dtype: int64

	calories	duration
0	420	50
2	390	45

A DataFrame is like a table with rows and columns. Pandas use the *loc* attribute to return one or more specified row(s). It returns a Pandas Series.

Read CSV

```
import pandas as pd

df = pd.read_csv('data.csv')

print(df)
print(df.to_string())
```

Useful Functions

- `dropna()`
- `info()`
- `isnull()`
- `apply()`
- `head()`
- `tail()`

DataFrames

```
import pandas as pd

data = {
    "calories": [420, 380, 390],
    "duration": [50, 40, 45]
}
```

```
df = pd.DataFrame(data)
print(df)
```

```
def make_double(x):
    return x * 2
```

```
df['calories'] = df['calories'].apply(make_double)
print(df)
```

Output:

	calories	duration
0	420	50
1	380	40
2	390	45

	calories	duration
0	840	50
1	760	40
2	780	45

Matplotlib

- Matplotlib is a low level graph plotting library in python that serves as a visualization utility.
- Matplotlib is open source and we can use it freely.
- Matplotlib is mostly written in python, a few segments are written in C, Objective-C and Javascript for Platform compatibility.

Using Matplotlib

```
C:\Users\Your Name>pip install matplotlib
```

```
# check matplotlib version  
  
import matplotlib  
print(matplotlib.__version__)
```

Matplotlib Pyplot

- Most of the Matplotlib utilities lies under the `pyplot` submodule, and are usually imported under the `plt` alias.

The `plot()` function is used to draw points (markers) in a diagram.

By default, the `plot()` function draws a line from point to point.

Matplotlib Pyplot

The `plot()` function is used to draw points (markers) in a diagram.

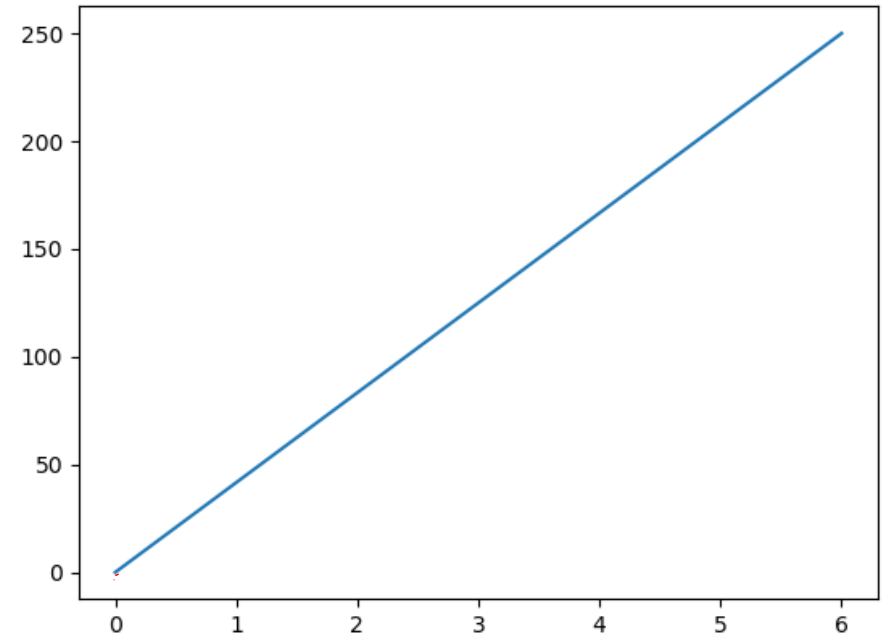
By default, the `plot()` function draws a line from point to point.

- Most of the Matplotlib utilities lies under the `pyplot` submodule, and are usually imported under the `plt` alias.

```
import matplotlib.pyplot as plt
import numpy as np
```

```
xpoints = np.array([0, 6])
ypoints = np.array([0, 250])
```

```
plt.plot(xpoints, ypoints)
plt.show()
```



References

- https://www.w3schools.com/python/python_getstarted.asp